

**Conifer Seedling Mapping with UAS Texture Analysis and Supervised Learning in Postfire
Southern Rockies Landscapes**

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Abstract

Accurate detection of seedling regeneration following disturbances such as wildfires is essential for monitoring ecosystem recovery, assessing forest resilience, and guiding restoration efforts. However, most conifer regeneration estimates in post-fire environments rely heavily on in-situ measurements, which are labor-intensive and spatially limited, due to the lack of spatial resolution in traditional remote sensing methods. Uncrewed Aerial Systems (UAS) equipped with very high-resolution red-green-blue (RGB) photography provide an innovative, low-cost solution for accurately detecting conifer seedlings. In this study, we leveraged UAS-collected RGB imagery to detect conifer seedlings in mountainous post-fire landscapes. Here, we define a seedling as a conifer tree above 30 cm and below 4 m in height. By integrating image texture, canopy structure, and RGB-based indices within a Random Forest framework, our model achieved 89% accuracy in conifer seedling detection. Detection accuracy was influenced by both seedling size and the spectral features used, with intra-class greenness variability emerging as the most influential factor. Additionally, incorporating 3D structural metrics, particularly seedling height enhanced detection accuracy by 10–15%. While textural variables help reduce detection uncertainty, higher overall accuracy can still be achieved without their inclusion. Our results demonstrate that UAS-derived RGB image texture can reliably distinguish conifer seedlings from shrubs and other vegetation across mountainous environments. By providing wall-to-wall estimates of seedling presence, UAS-based detection overcomes the spatial limitations of field-based measurements and offers a scalable approach to validate and complement coarser remote sensing data. These findings reinforce the value of UAS as a powerful tool for post-fire forest monitoring, offering high-resolution, scalable data critical for guiding restoration and adaptive management efforts.

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47 **Keywords:** Uncrewed Aerial Systems, Conifer Seedlings, Machine Learning, Wildfire, Post-fire
48 regeneration

49 **Highlights**

- 50 • High-resolution RGB sensors offer a low-cost, scalable method for detecting conifer
51 seedlings.
- 52 • Integration of RGB indices and structure achieved ~ 89% detection accuracy.
- 53 • Inclusion of seedling height improved detection accuracy by 10–15%
- 54
- 55

1. Introduction

Accurate, reliable, and cost-effective methods of evaluating forest regeneration following disturbances are needed to improve forest inventories, management, and silvicultural operations. This need is especially acute in the context of climate-driven shifts in disturbance regimes of the Western United States, including larger and more intense fires (Abatzoglou & Kolden, 2013) and increased prevalence of drought and pests (Bentz et al., 2010). Natural regeneration of conifer forests following these disturbances is no longer guaranteed, threatening conifer forest persistence (Davis et al., 2023a). Rapid and accurate quantification of natural ecosystem recovery across large areas is essential for making informed decisions regarding forest management in the face of this threat (Rodman et al., 2024). While advances in remote sensing now allow for broad-scale assessment of long-term ecosystem recovery, there is a documented need to understand the dynamics of natural seedling recruitment and growth in initial stages of succession (Davis et al., 2023a).

Restoration and reforestation interventions also require accurate, reliable, and cost-effective methods of monitoring the natural regeneration, and survival and new establishment of planted tree seedlings in restored areas (Fargione et al., 2021; North et al., 2019). However, satellite imagery is unable to capture seedling dynamics, as seedling spectral characteristics are very much similar to those of co-existing shrubs. Moreover, early postfire years are hard to assess because rapid “green-up” in optical indices only often reflects shrub resprouting rather than conifer seedling recovery, so greenness can overstate forest return (Bright et al., 2019; Celebrezze et al., 2024) (Figure 1). These challenges lead most early forest recovery research and monitoring programs to rely on costly field surveys involving manual seedling counts (e.g., Marshall et al., 2022). With increases in disturbed areas across the Western United States,

adequate monitoring of seedling dynamics across environmental gradients through in-situ methods has become difficult with limited funding and capacity. Land managers need accurate and timely seedling recovery data in order to best prescribe restoration treatment. Additionally, comprehensive information on spatial coverage, physical location, and structural characteristics of seedlings, which are difficult to obtain by traditional field methods, are critical to understanding natural regeneration strategies and ecosystem resilience.

Recent studies have leveraged high-resolution imagery from Uncrewed Aerial Systems (UAS) to overcome the limitations of forest monitoring associated with both satellite-based remote sensing and manual methods (Marcial-Pablo et al., 2019; Schiefer et al., 2020). However, there is a lack of studies on utilization of quantitative seedling detections in post-disturbance environments, especially across varied terrain. While expensive specialized UAS with Light Detection and Ranging (lidar) and hyperspectral sensors can provide highly-detailed information about vegetation type and structure, more conventional red-green-blue (RGB) image-based UAS can effectively capture information about small vegetation structures at high spatial resolution and low cost, expanding potential management applications (Dainelli et al., 2021). By implementing a faster region-convolutional neural net (R-CNN), Fromm et al., 2019 detected conifer seedlings at 81% precision along seismic lines and showed that precision varied with training-set size, season, seedling size, and spatial resolution. However, neural nets require significant training data. In other cases, researchers have delineated individual vegetation and their structural metrics, followed by a classification utilizing structural attributes (e.g., canopy height, point density, foliage height diversity) (Cunliffe et al., 2016). While these structural attributes are effective for tree classification, they may lead to higher inaccuracies when distinguishing seedlings, which can easily be confused with shrubs at peak greenness (Feduck et al., 2018). The

use of vegetation indices derived from RGB bands provide additional information and can increase the classification accuracy of crops and short height vegetation (Zhang et al., 2021). The optical data respond strongly to leaf area rather than vertical structure, spectral recognition of vegetation types can diverge from true types showing a difficulty in resolving finer details of shrubs and seedlings if used alone (Anees et al., 2025; Ayhan & Kwan, 2020; Zahabnazouri et al., 2025). Further, classification accuracy depends on the growth stages, and thus, there is a need for multiple indices that can capture variability of the same vegetation to improve accuracy (Nguy-Robertson et al., 2012). However, the spatial variability and patterning of individual pixels (aka:Texture) from each band have the potential of identifying finer scale variations between shrubs and seedlings. Texture variables have been commonly used in coarser land cover classification due to their proven ability to increase classification accuracy (Rodriguez-Galiano et al., 2012). The use of texture variables in UAS-based image vegetation classifications is gaining attention; recent work demonstrates that textures can differentiate surfaces at a finer level, increasing the number of distinguishable classes (e.g., different types of vegetation) and enhancing class separability and classification accuracy (Gini et al., 2018). For example, the information added from single-band texture variables is useful for compensating for difficulties of performing multi-temporal surveys and can significantly improve the accuracy of classification (Coburn & Roberts, 2004; Gini et al., 2018). In precision agriculture, texture measures derived from multispectral UAS imagery improved weed detection amidst crops (David & Ballado, 2016). However, use of texture variables to detect seedlings across a range of vegetation densities and topographic gradients in post-fire environments has not been tested.

Much of the research on UAS-based textural classification neither investigates vegetation smaller than 1 m nor classifies sparse and small vegetation, which is often the situation in some post-disturbed ecosystems, such as dry forests. Further, there is minimal research on utilizing these texture, structure, and vegetation indices together in post-fire seedling detection in conifer systems. In this study, we test the effectiveness of high-resolution RGB UAS imagery and machine learning techniques to spatially map conifer seedlings defined as the new conifers generated post-fire and surrounding other major vegetation types including shrubs, standing dead trees, and mature unburned trees in mountainous terrain in the Southern Rockies ecoregion. We provide a detailed analysis on the best remotely sensed variables for seedling detection in mountainous landscapes, the seedling and site-specific characteristics that can improve detection performance, and insights on scaling of such measurements to inform monitoring of large disturbances across ecosystem and elevation gradients.

2. Materials and Methods

2.1 Study area

Our research was conducted in wildfire-impacted ponderosa pine (*Pinus ponderosa*) and Douglas-fir (*Pseudotsuga menziesii*)-dominated forests across the Colorado Front Range in the Southern Rockies ecoregion of the United States (Figure 2). These mixed-conifer forests are generally constrained by elevational gradients to the montane zone from 1,800 m to ~3,000 m (Huckaby et al., 2003; Sherriff & Veblen, 2007) and are soil moisture-limited (Oliver & Ryker, 1990). They are often characterized by open, park-like stands of ponderosa pine on southern aspects and xeric or lower elevation sites near the grassland/shrubland ecotone, with more dense,

mixed-conifer stands of ponderosa pine and Douglas-fir on northern aspects and mesic or higher elevation sites (Benedict, 1991; Huckaby et al., 2003; Marr, 1961; McKinney, 2019; Sherriff & Veblen, 2007). Rocky Mountain juniper (*Juniperus scopulorum*) is another common associate, especially at lower elevations as these forests transition to open ponderosa savannas and shrublands (Benedict, 1991; Huckaby et al., 2003). At the upper elevational boundary of the montane zone, ponderosa forests become intermixed with and eventually transition to subalpine forests consisting of quaking aspen (*Populus tremuloides*), blue spruce (*Picea pungens*), subalpine fir (*Abies lasiocarpa*), Engelmann spruce (*Picea engelmannii*), limber pine (*Pinus flexilis*), and lodgepole pine (*Pinus contorta*) (Chambers et al., 2016; Huckaby et al., 2003; McKinney, 2019).

We selected four specific burn scars for this study from the Monitoring Trends in Burn Severity (MTBS) dataset: the Big Elk Fire (2002), Schoonover Fire (2002), Hayman Fire (2002), and High Meadows Fire (2000) (Eidenshink et al., 2007). These fires were selected for their location on non-wilderness United States Forest Service (USFS) lands and composition of over 50% ponderosa pine and mixed-conifer montane forest as defined by the Landfire Biophysical Settings dataset (Rollins, 2009). These sites are additionally characterized by varied topography and seedling densities with a range of shrub and mature tree densities to assess the transferability of our methods (Figure 2 B, Figure 2C). Selected fires occurred 20-23 years prior to our research, allowing enough time for seedlings of various sizes and growth stages to naturally emerge without any restoration efforts across the landscape (Figure 3). To our knowledge, none of the selected sites were artificially seeded following wildfire. This variability allowed us to

evaluate the marginal seedling size that could be detected using high-resolution, low-cost UAS with reduced training data.

2.2 Data Collection & Processing

The overall seedling detection workflow (Figure 4 and Figure 5) has three main steps; 1) initial Structure-from-Motion (SfM), 2) vegetation crown delineation, 3) classification. We collected RGB images over 11 seedling census plots using a DJI Phantom 4 Pro drone during the summer of 2022. UAS image collection parameters can be found in supplementary table 1. Sixty-meter diameter census plots were non-randomly selected in order to ensure topographic and land cover variation, both with and without high densities of regenerating seedlings with a range of heights (Table 1). We also established these plots primarily in moderate to high severity patches within each fire. The fire severity was calculated using modeled bias-corrected Composite Burn Index (CBI) (Parks et al., 2019) derived using Landsat imagery and Google Earth Engine (GEE) (Gorelick et al., 2017) according to methods from Parks et al. 2019. All UAS mission planning and execution was performed using Map Pilot Pro software (v23) and the DJI Phantom 4 Pro drone. Data collection occurred in clear and scattered cloud conditions with minimal wind conditions (0-5 mph) using a high photo density and along/across-track overlap, as well as terrain-following capabilities (see Supplementary Table 1 for flight specification and unit details). The final Ground Sampling Distance (GSD) of all the flights was within 0.7 - 0.8cm. All images were collected using WGS 84 global geographic projection and used the built-in drone unit GPS system for navigation.

We used Agisoft Metashape software (v1.8.1) to pre-process all UAS imagery and generate RGB orthomosaics and high-resolution point clouds from the RGB images using the University of

Colorado Boulder High Performance Computing (HPC) system. A detailed processing parameters can be found in the supplementary materials (Supplementary table 2). Following image processing, seedling censuses were performed across the selected plots to train and test a seedling detection model. Seedling censuses were performed using the digital mapping procedure from Graves et al., 2018. The pre-processed RGB orthomosaics for each plot were turned into tile packages (.tpkg) via ESRI ArcMap and loaded onto ArcGIS Fieldmaps, software created for geographic data collection in the field. Field crews mapped spatially explicit polygons of all seedlings, shrubs, trees, and standing dead trees onto the pre-processed RGB orthomosaics, each with 3 cm spatial resolution. We measured seedling heights using a TruPulse 360R laser rangefinder when over ~1 m in height and a manual tape measure when less than ~1 m in height. We recorded all the seedlings and shrubs taller than 0.3 m due to grass cover up to 30 cm obscuring shorter seedlings. In total, seedling censuses identified 663 seedlings, ranging in height from 0.3 m to 4.3 m (Table 1). While not traditionally considered a seedling, we included vegetation 4.3 m or less in the seedling class because it would have sprouted after the fire, and thus, been a seedling at some point in the last two decades. In addition to these individual vegetation surveys, we opportunistically collected polygons from each plot representing larger established trees (evergreen and deciduous), herbaceous (grass/moss/herbs), rock, shadowed regions, and bare ground. In our classification model, we included all eight classes.

Table 1: In-situ field census used in this project. All the aboveground woody vegetation was surveyed from 10 plots with approximate size of 2817 m² (60 m diameter circular plots).

Fire	Area surveyed (m²)	Slope	Seedlings (evergreen)	Mature (evergreen)	Shrubs	Deciduous	Standing dead
Hayman	2817	Flat	305	0	0	9	4
High Meadows	2829	Moderate	47	1	18	0	0
High Meadows	2817	Flat	149	0	11	0	0
High Meadows	2817	Flat	6	0	142	0	1
Big Elk	2817	Steep	52	1	15	3	12
Big Elk	2817	Steep	13	15	20	4	1
Schoonover	2817	Moderate	21	0	68	0	24
Schoonover	2817	Flat	20	16	41	0	10
Schoonover	2817	Moderate	22	1	28	3	3
Schoonover	2817	Moderate	0	1	0	0	0

2.3 Individual tree delineation and point cloud parameters

Three-dimensional point clouds for each census plot were derived using SfM processing in Agisoft Metashape. Point clouds were exported and used in R statistical software (R Core Team, 2024 v4.3.2) to generate a digital terrain model (DTM), canopy height model (CHM), individual vegetation crowns, and their potential canopy height and canopy area (Figure 5B, C, and D). Point cloud-based data processing was implemented using the lidR package (Roussel et al., 2020). The digital terrain and canopy height models were derived at 0.1 m spatial resolution. Any vegetation above 0.3 m with more than 5 spatial points around the center point was delineated into crowns. Vegetation center point and crown delineation were done using the local maximum filter (lmf) algorithm using the lidR package in R (Roussel et al., 2020). Canopy heights above 4.3 m were excluded based on the maximum height of the seedling in our census data was ~4.3 m. We then used the CHM to derive mean canopy height and rumple (the randomness of vegetation height distribution within a given space) index variables within the delineated tree crowns (Figure 5 E).

2.4 RGB vegetation indices and texture variables

We used the 3 cm resolution orthomosaic with three bands (red, green, and blue) to calculate a set of RGB-based vegetation indices (Supplementary Table 3). The selection of vegetation indices were based on their ability to differentiate vegetation types in previous classifications on range of trees, grass, and shrub vegetation environments (Wan et al., 2018; Zhang et al., 2021). We generated 13 vegetation index variables. We further developed texture variables from each red, green, and blue band, helping to aggregate unique features based on reflectance distribution (Figure 5). We applied a 9-pixel moving window on the red, blue, and green reflectance bands to calculate the mean, variance, homogeneity, entropy, contrast, dissimilarity, correlation, and

second moment. This step generated 24 texture variables (eight from each band). The description of each variable can be found in the supplementary table 3. We then extracted the means of both texture and vegetation index variables within the field census polygons. Sample of index and texture variables showing the same in-situ plot is shown in figure 6.

2.5 Seedling detection model training and validation

We implemented three different machine-learning models to classify vegetation in our UAS imagery into four major aboveground classes of standing dead, deciduous, seedlings, mature trees (evergreen), and shrubs. These models (random forest (RF), support vector machine (SVM), and neural network (*nnet*)) were trained and validated using 38 variables derived from the texture, vegetation indices, CHM, and point cloud. Here, we report detailed methods for the random forest classifier, which performed best and was used for all further analyses. Details on the implementation of our support vector machine, and neural network models can be found in the Supplementary materials. We applied a random forest following Breiman, 2001, which was implemented using a cross-validated Random Forest model for multi-class classification using the caret package in R (Kuhn, 2008). This methodology uses stratified k-fold cross-validation to ensure robust performance estimation. We randomly selected 50% of the polygons for model training, which was then partitioned into 10 folds, ensuring balanced class distribution. We used a list of random seeds generated for each iteration to maintain reproducibility, with an additional seed for the final model. We performed grid search hyperparameter tuning while computing class probabilities and trained a final model using these parameters. We subsequently assessed the importance of features and executed a cross-validation process. This structured approach enhances model robustness, reduces overfitting, and optimizes hyperparameters, ensuring an

efficient and accurate predictive model. After these training steps, we assessed the importance of variables and the accuracy change with the addition of the number of variables. Our results suggest that a maximum of eight variables produces the highest stable results in this context. Using only the eight most important variables, we then re-ran our Random Forest model. The accuracy of the seedling detections are summarized as the F score (Equation 3) as it incorporates both the true and false positive detections calculated using Recall and Precision (Equations 1 and 2). To further assist with feature diagnostics, we plotted Area under curve (AUC), partial dependency plots (PDP), and Individual Conditional Expectation (ICE) plots from the random forest outcomes. These diagnostics can help in cases where feature importance is very similar for several classes.

$$\text{Recall} = \frac{\text{\# of ground seedlings detected by drones}}{\text{\# of ground seedlings}} \text{----- (1),}$$

$$\text{Precision} = \frac{\text{\# of ground seedlings detected by drones}}{\text{\# of drone detected seedlings}} \text{----- (2),}$$

$$\text{F score} = \frac{\text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}} \text{----- (3),}$$

3. Results

3.1 Detection of seedlings in mountainous terrains

Using Random Forest (RF) classification algorithm, we were able to detect seedlings in post-fire environments of the Southern Rockies with 89% overall accuracy across all plots surveyed regardless of seedling and other vegetation feature densities and site characteristics such as variable slope and elevation (Table 2). Initially we tested three methods used for similar types of classifications in literature and found that RF performed the best, hence RF based results only are presented in the main manuscript. Table 2 and figure 7 shows the comparison of each algorithm tested for classification performance. Details of SVM and *nnet* methods can be found in the supplementary information.

Table 2: Performance metrics from the three main classifiers used (random forest, Support vector machine, and neural network). The random forest out performs the other two classification models and is used for further processing.

Performance metric	Random forest (rf)	Support vector machine (svm)	Neural network (nnet)
Accuracy	0.89	0.80	0.78
Precision	0.91	0.57	0.56
Recall	0.91	0.50	0.41
F score	0.91	0.52	0.42

Across fires, our field census data showed a range of seedling and other vegetation densities, growth forms, and sizes (Figure 2C). Seedling density varied from 1- 305 seedlings per plot with a range of densities of other vegetation types collected across the 10 plots we surveyed (Table 1). Seedling heights varied within and across plots (0.3 m to 4.3 m). The other standing vegetation types such as shrubs, deciduous trees, and standing dead trees also varied in count and distribution across and within plots (Figure 2C and Table 1). Mountainous terrain makes post-disturbance analysis of vegetation more challenging due to the remoteness and terrain-related access difficulties. These regions are further impacted by increased soil erosion and post-disturbance climate and vegetation invasions. Our study showed that the use of UAS images derived variables can accurately and quantitatively detect seedlings as small as 30 cm in height and distinguish them from similar other vegetation such as shrubs in variable slopes and vegetation densities. Samples of three census plots with census polygons and post classified data are shown in figure 8.

3.2 Classification of aboveground vegetation using machine learning

Our results confirmed that the RF can detect conifer seedlings across topographic gradients at high accuracy (Table 2). The overall importance and class base importance of variables in the classification model are shown in figure 9. Across all spectral, structural, and textural features, only a few features stand out as contributing significantly to differentiation of the five major vegetation classes investigated. Our results showed that the variability in canopy height (Variance CHM) is the most effective variable to separate vegetation states, consistent with expectations that crown relief and height heterogeneity can distinguish woody canopies from low or sparse cover (Figure 9 B). We further found that Greenness and related indices (Excess Green

316 Index (EXG), Green Index (GI), Green Red Vegetation Index (GRVI), and Modified Green Red
317 Vegetation Index (MGRVI)) are particularly informative for separating Seedlings and Shrub
318 classes (Yellow in high heatmap scores) reflecting the effectiveness of capturing differences in
319 chlorophyll intensity and red-green contrast typical of actively photosynthesizing, short
320 vegetation (Figure 9 B and C). Our method enhanced the detection of shrubs, the major
321 vegetation class that can easily be mixed with the seedlings, as observed in the field surveys and
322 literature (Ayhan & Kwan, 2020).

323 The most important variable for separating mature conifer trees from young postfire seedlings
324 were the canopy height and its variation. The importance of green indices for classifying mature
325 conifer trees is minimal, indicating that structural texture (variance CHM) carries more
326 discriminatory power than color alone in tall, closed canopies. Deciduous trees showed moderate
327 reliance on green based indices as of shrubs, suggesting leaf chemistry and phenology captured
328 by red, green, and blue relations helps in differentiation of woody lifeforms of similar stature. In
329 contrast, dead vegetation showed high importance for EXG and related greenness metrics
330 reflecting the reduced and absent chlorophyll with modest contribution from Variance CHM,
331 which was the most important for mature trees (these dead trees are pre-fire mature trees burned
332 during the fire).

333

334 We found that EXG (Excess Green) and GLA (Green Leaf Area Index) were more useful for
335 dead and seedling vegetation detection. EXG's high importance for dead vegetation may indicate
336 strong contrast between brown or dead materials and any remaining green within them. GLA
337 distinguishes deciduous types with higher seasonal green reflectance variability. Our study
338 conducted during peak greenness season considered the accessibility to most of these study sites

339 and potentially for other similar mountainous sites to provide the best estimates for seedling
340 detection. However, our results showed that collection of UAS images across sessions including
341 the leaf off season where deciduous shrubs and trees can easily be detected from evergreen trees
342 can enhance the accuracy of classification. We further noticed that texture variables did not stand
343 out within the first eight most important variables, however, they appeared within the first 12
344 variables. The first twelve variables including the texture variables provided the same accuracy
345 as with eight variables (89%), however with lower uncertainty than that with eight variables. Our
346 further investigation demonstrated that the same variables were equally important distinguishing
347 a few classes (Figure 10).

348 The Area Under the Curve (AUC) lollipop plots showed the probability of each feature for
349 detecting classes selected in this study (Figure 10). Shrubs and seedlings showed very similar
350 probability in each variable for separating them from each other. However, these two classes
351 showed very distinct patterns in the partial dependency plots (PDP) and the Individual
352 Conditional Expectation (ICE) plots (Figure 11). For example, Seedlings (Figure 11 A) showed
353 that the predicted probability increases sharply when EXG exceeds ~25, suggesting higher
354 seedling presence in areas with strong green dominance. Figure 11 B shows the probability of
355 where shrub classification declines rapidly as GRVI increases above 0, indicating that low or
356 negative GRVI values (associated with senescent or woody vegetation) are more indicative of
357 shrub cover. In the case of mature trees (Figure 11 C, variance in canopy heights strongly
358 influences classification, with probabilities increasing steeply as structural variability exceeds ~2
359 m, reflecting the heterogeneous vertical structure of mature tree canopies. The standing dead
360 (Figure 11 D showed that its probability drops substantially when GI exceeds ~0.34, consistent
361 with the spectral signature of dead vegetation lacking green biomass.

Together, the AUC rankings and ICE/PDP behavior demonstrated our RF classifier as a unique and effective way to classify the confusing classes in a postfire environment using structure and spectra based minimal feature sets. For example, our classifier used Variance CHM to anchor tree detection and a single robust greenness proxy (EXG/GRVI) to separate seedlings/shrubs from dead (Figure 11). This aligned with the broader literature recommending fusion of lidar derived structure and optical indices for vegetation type and condition mapping (Dalponte et al., 2008).

We further investigated the importance of fire, slope, and other environmental parameters in differentiating those classes of interest in this study. However, we found that those classes did not have significant importance and thus, we show that with spectral and structural variables we can detect the conifer seedlings across a range of topographic and vegetation density conditions in mountainous terrains.

4. Discussion

In this study, we assessed the capability of detecting conifer seedling presence in mountainous postfire ecosystems using UAS-derived variables and machine learning classifiers, including Random Forest, Support Vector Machine, and Neural Networks. Integrating vegetation structural, textural, and biophysical attributes improved discrimination of conifer seedlings from structurally similar vegetation, particularly co-occurring shrubs. Predominant species included ponderosa pine (*Pinus ponderosa*) and Douglas-fir (*Pseudotsuga menziesii*) and various mountainous shrub species and deciduous trees including quaking aspen (*Populus tremuloides*). Comparisons across all three models revealed that classification performance was predominantly influenced by the choice of predictor variables rather than the model architecture itself (see Supplementary Materials). Specifically, our study reveals that structure and spectra-based

variables are most effective when classifying vegetation from the data collected during peak greenness period.

Random Forest demonstrated higher accuracy (89%) across models tested and computational efficiency for this task and was selected as the choice model approach. We demonstrated that using RGB spectra and key structural features derived from UAS images (e.g., CHM, EXG, and several other greenness-based indices), we can successfully (with 89% accuracy) recognize conifer seedlings as small as 30 cm in height. Further, this delineation was dependent on variable densities of other vegetation, specially from similar sized shrubs. This methodology will advance conifer seedling detection at scale and time (based on seedling sizes), which is critical for assessing post-disturbance forest regeneration and long-term ecosystem resilience. Conifer seedlings are particularly vulnerable in the early years following disturbances such as wildfire, drought, or logging, making their detection essential for evaluating recovery trajectories (Andrus et al., 2022; Davis et al., 2023b; Shepperd et al., 2006). Traditional field surveys provide detailed species-level data but are labor-intensive and limited in spatial extent. Furthermore, access to most post-disturbance mountainous landscapes is limited to the summer months, due to persistent snow cover from late spring melt and early autumn snowfall. In addition, there is no such study that detects conifer seedlings of very short height across spatial scales only using data collected during peak greenness, which is essential for post-disturbance ecosystem recovery and resilience.

However, challenges remain in differentiating conifer seedlings from other early successional species such as woody shrubs, especially in heterogeneous post-disturbance mountainous landscapes. The ability to accurately detect conifer seedlings is essential for forecasting forest regeneration under shifting disturbance regimes and climate change. In many temperate and

408 boreal forests, conifer dominance relies on successful seedling establishment within a specific
409 post-disturbance window, often constrained by moisture availability, seed source proximity, and
410 competition from shrubs and grasses (Roccaforte et al., 2012; Stevens-Rumann et al., 2018).
411 Failure to detect insufficient regeneration early may lead to missed management interventions
412 and permanent ecosystem transitions, such as shifts from conifer forests to shrublands or
413 grasslands. Quantifying seedling presence and spatial patterns allows land managers to prioritize
414 replanting, control competing vegetation, and assess the effectiveness of post-fire rehabilitation
415 efforts. Integrating seedling detection into disturbance monitoring frameworks enhances
416 predictive models of forest succession and informs adaptive management strategies aimed at
417 sustaining forest structure and function under increased disturbance pressures.

418 Future improvements to this workflow could utilize a time-series and multispectral approach
419 accounting for vegetation phenology and ecosystem state changes (e.g. conversion from forest to
420 grassland). In conifer-dominated systems, evergreen canopies undergo seasonal physiological
421 and soil moisture-driven changes (e.g. photosynthetic pigments and foliar water content) (Gamon
422 et al., 2016; Lad et al., 2023). These shifts can alter reflectance and textural properties,
423 potentially causing classification confusion between shrubs and conifer seedlings (Dronova &
424 Taddeo, 2022). Transitions in ecosystem state driven by invasive annual grasses are also
425 important to capture when assessing ecosystem recovery. These transitions can be more
426 accurately detected when incorporating multispectral time series, which account for earlier
427 greening of invasive grass species (Kreitler et al., 2025). For sites where ecosystem transitions
428 are a management concern, revisiting over the course of the growing season and detection
429 utilizing multispectral imagery could further classify annual grass, shrubs, and conifer sapling.

430 Despite the demonstrated accuracy of this workflow, UAS-based approaches remain data-
431 intensive and computationally demanding. High-resolution image acquisition, photogrammetric
432 reconstruction, and feature extraction generate large datasets that require substantial storage,
433 processing power, and workflow expertise. Our study highlights the utility of the cloud-based
434 computational infrastructure CyVerse to streamline data management, house reproducible
435 workflows, and facilitate collaboration (Goff et al., 2011; Merchant et al., 2016; Swetnam et al.,
436 2024). These challenges mirror recognized efforts to bring UAS into ecological monitoring,
437 emphasizing the need for standardized, open, and scalable workflows (Koontz et al., 2022).
438 Wildfires are one such inevitable disturbance and show increased frequency, severity, and
439 growth over the last two decades in the western US conifer forest systems (Balch et al., 2024;
440 Parks & Abatzoglou, 2020; Williams et al., 2022)). With this trend, post-fire ecosystem
441 management has been taking a variety of steps to enhance vegetation regeneration and
442 restoration with a variety of implementations to avoid unnecessary ecosystem transformations.
443 Successful implementation of post-fire ecosystem management requires accurate quantitative
444 evaluation of regeneration ability under variable post-fire climate and site conditions. In this
445 study, we selected a range of fires in the Southern Rockies ecoregion and collected UAS based
446 image data and implemented a machine learning based approach on vegetation indices, texture
447 and structural features to detect the conifer seedlings as small as 30 cm in height. We further
448 assessed the detection success when presenting other vegetation such as woody shrubs,
449 deciduous trees and standing dead trees. In addition, we assessed some site specific and
450 topographic properties on detecting conifer seedlings from other major vegetation classes in a
451 post disturbance environment. Our results show that only using spectral and structural features
452 derived from UAS images, we can successfully (89% accuracy) recognize conifer seedlings as

453 small as 30 cm in height. However, due to both the expanse of public lands, limited accessibility,
454 and complex topography, evaluation of post-fire environments is extremely challenging. Surový
455 et al., 2018 described limitations in detecting small seedlings using UAV-derived models,
456 especially of seedlings with low heights. Temporal mismatch in data collection and limited
457 ground truth for training can also introduce error (Getzin et al., 2014; Stevens-Rumann et al.,
458 2018). These challenges emphasize the importance of detecting variable features and site
459 settings, and ground validation in improving detection accuracy. Our study, derivations of a
460 range of metrics from vegetation indices, texture, and structural metrics, allowed us to test the
461 seedling detection overcoming the challenges posed in previous studies.

462 This research is especially relevant given the context of expansive reforestation efforts across the
463 United States. In 2021, the REPLANT Act directed and funded the USFS to plant over 1 billion
464 trees in the next decade (USDA Forest Service, 2022). Prior to this act, USFS resources were
465 only able to address ~6% of post-wildfire reforestation needs annually (USDA Forest Service,
466 2022). There are also substantial efforts also at the state level to assess the postfire impact and to
467 intervene postfire conifer restoration and managing (Marshall et al., 2024; Rodman et al., 2025).

468 With significant public resources now being devoted to the national reforestation pipeline
469 (Fargione et al., 2021), broad-scale seedling monitoring—such as that enabled by the methods
470 used here, will be essential to ensure effective use of resources. This work is especially relevant
471 as Ponderosa Pine is the most commonly planted tree species in the western U.S. (Fargione et al.,
472 2021). Overall, machine learning approaches offer substantial improvement for detecting conifer
473 seedlings in post-fire landscapes from airborne observations. This offers the potential for a more
474 cost-effective and efficient means to take stock of western US forests and how they are
475 recovering post-disturbance.

476

477 **5. Conclusion**

478 This study demonstrates the efficacy of UAS-derived high-resolution RGB imagery combined
479 with machine learning for detecting postfire conifer seedling regeneration across mountainous
480 landscapes. By integrating spectral indices, spatial texture, and structural metrics, we achieved a
481 robust classification framework, with Random Forest yielding greater than 89% accuracy.
482 Vegetation structure and greenness variations and their correlation emerged as strong predictors,
483 enabling reliable differentiation between seedlings and shrub vegetation. Although the inclusion
484 of textural variables reduces detection uncertainty, they are not essential for achieving higher
485 classification accuracy. This approach provides a scalable, low-cost alternative to labor-intensive
486 field surveys, offering a transformative tool for precision forest monitoring, ecosystem recovery
487 assessments, and climate-resilient land management in remote mountainous regions.

488

489

490 **Data and Code Availability Statement**

491

492 The raw data including UAS images and metashape outputs can be found at
493 10.5281/zenodo.17457807. The codes and final results can be found at
494 [Conifer_seedling_mapping](https://github.com/ConiferSeedlingMapping/Conifer_seedling_mapping) GitHub repository.

495 **Conflict of Interest**

496 The authors declare that the research was conducted in the absence of any commercial or
497 financial relationships that could be construed as a potential conflict of interest.

498 **Author contributions**

499 Author contributions to this article are outlined using the CRediT contributor role taxonomy
500 (Brand et al., 2015).

- 501 • Writing - Original Draft: NTI, TLM
- 502 • Writing - Review and Editing: All authors
- 503 • Conceptualization - NTI, TLM, JKB
- 504 • Project Administration - TLM, NPH
- 505 • Visualization - NTI, TLM, NPH
- 506 • Supervision - JKB, TLM, NTI
- 507 • Funding Acquisition - JKB
- 508 • Methodology and Analysis - NTI
- 509 • Data Management - NTI, TLM
- 510 • Data Collection - NTI, TLM, KE, SSS, IM

511

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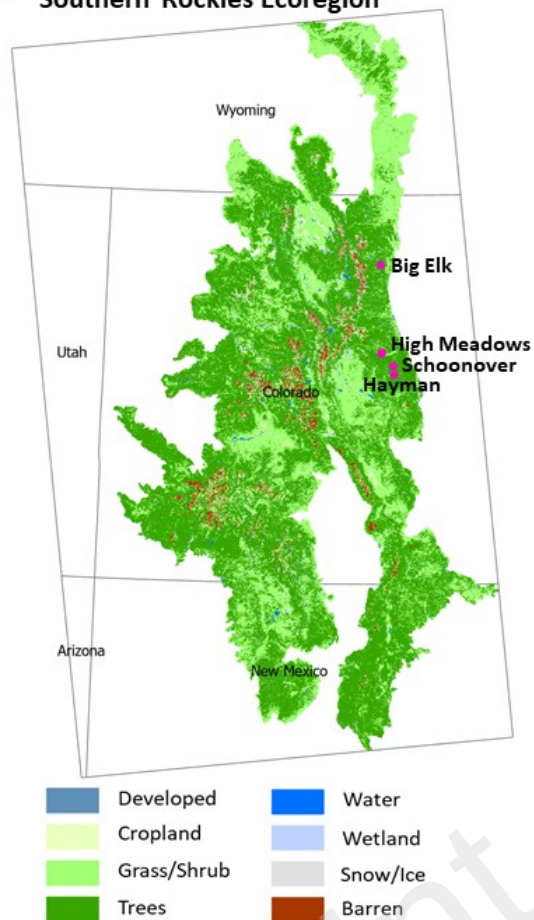
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A Southern Rockies Ecoregion

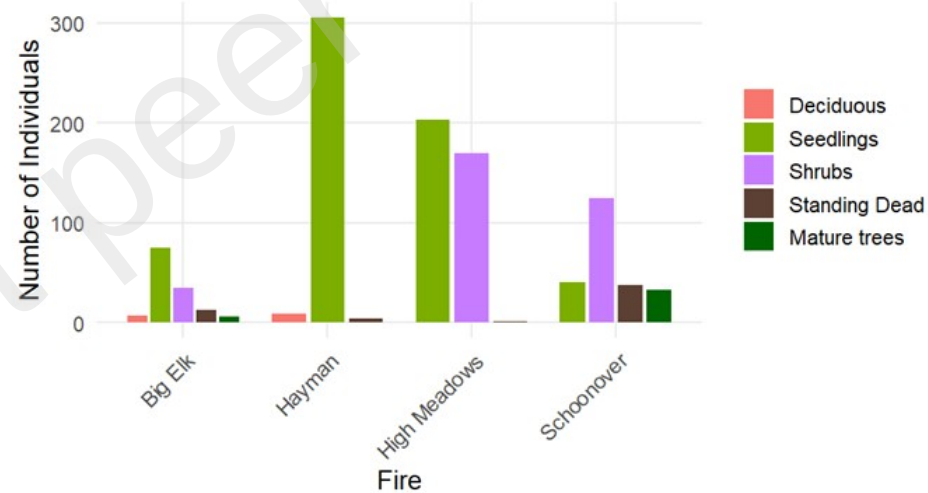


B



C

Frequency of Vegetation Classes by Fire





Seedling and shrub combination with dense standing dead trees



Aspen succession after fire

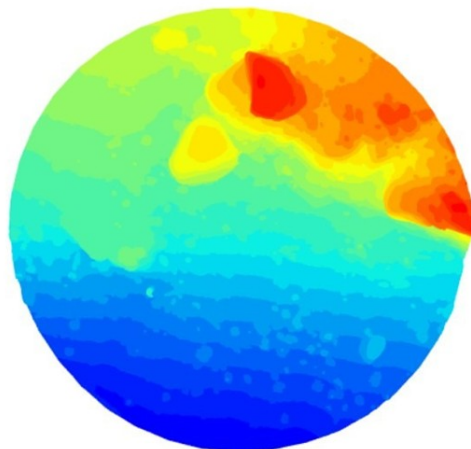


UAS image collection over a burned area





Vegetation and topography of a plot in RGB colors



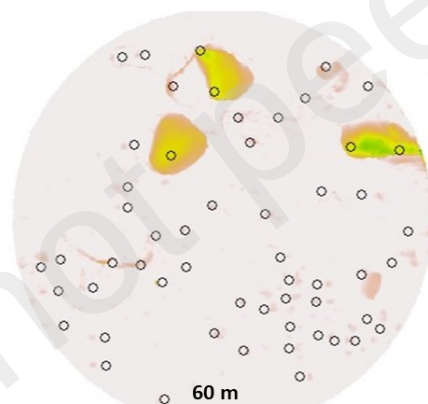
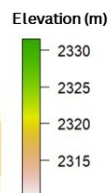
Vegetation and topography distribution as a 3D point cloud



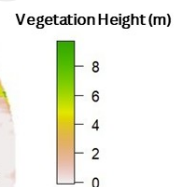
Classified point cloud with blue as ground and white as any vegetation

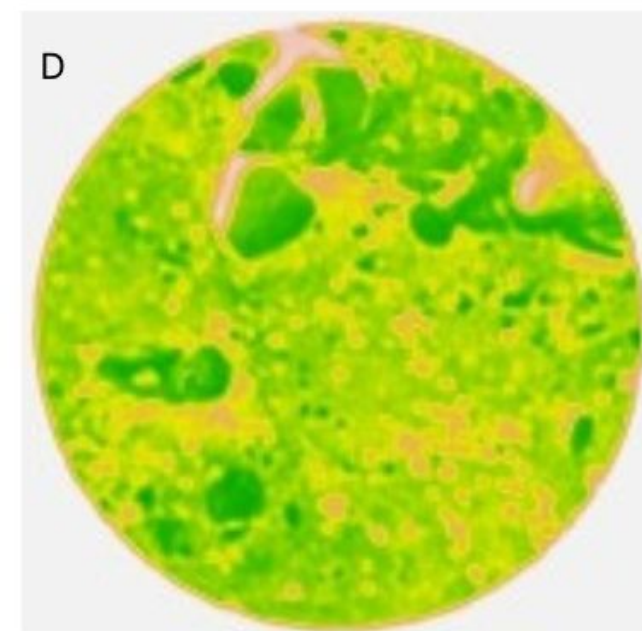
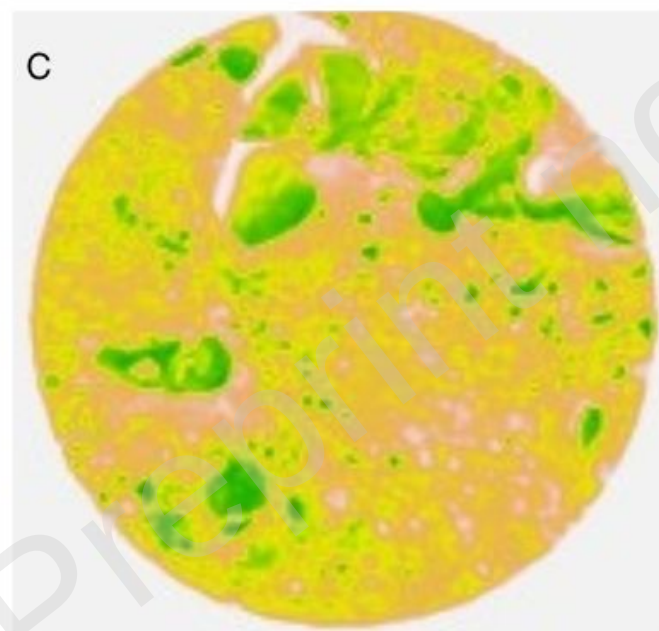
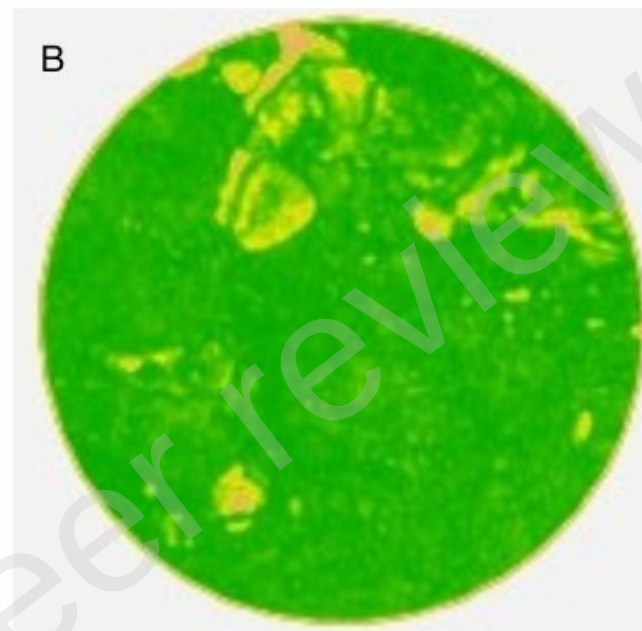
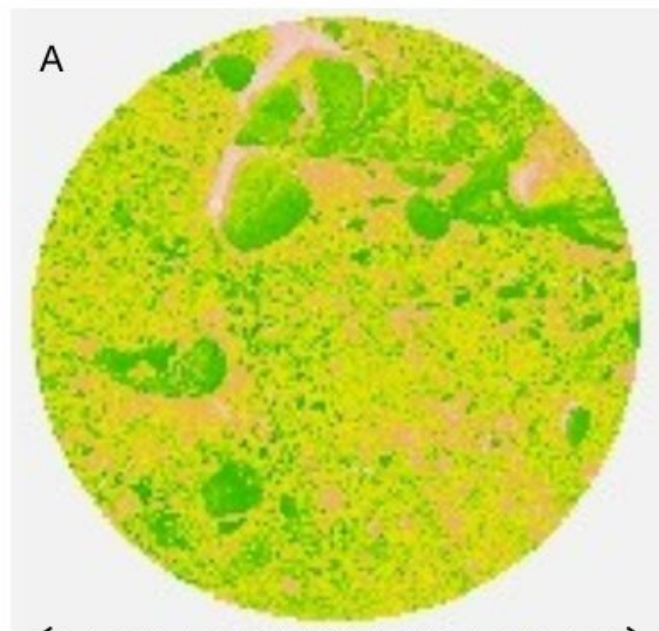


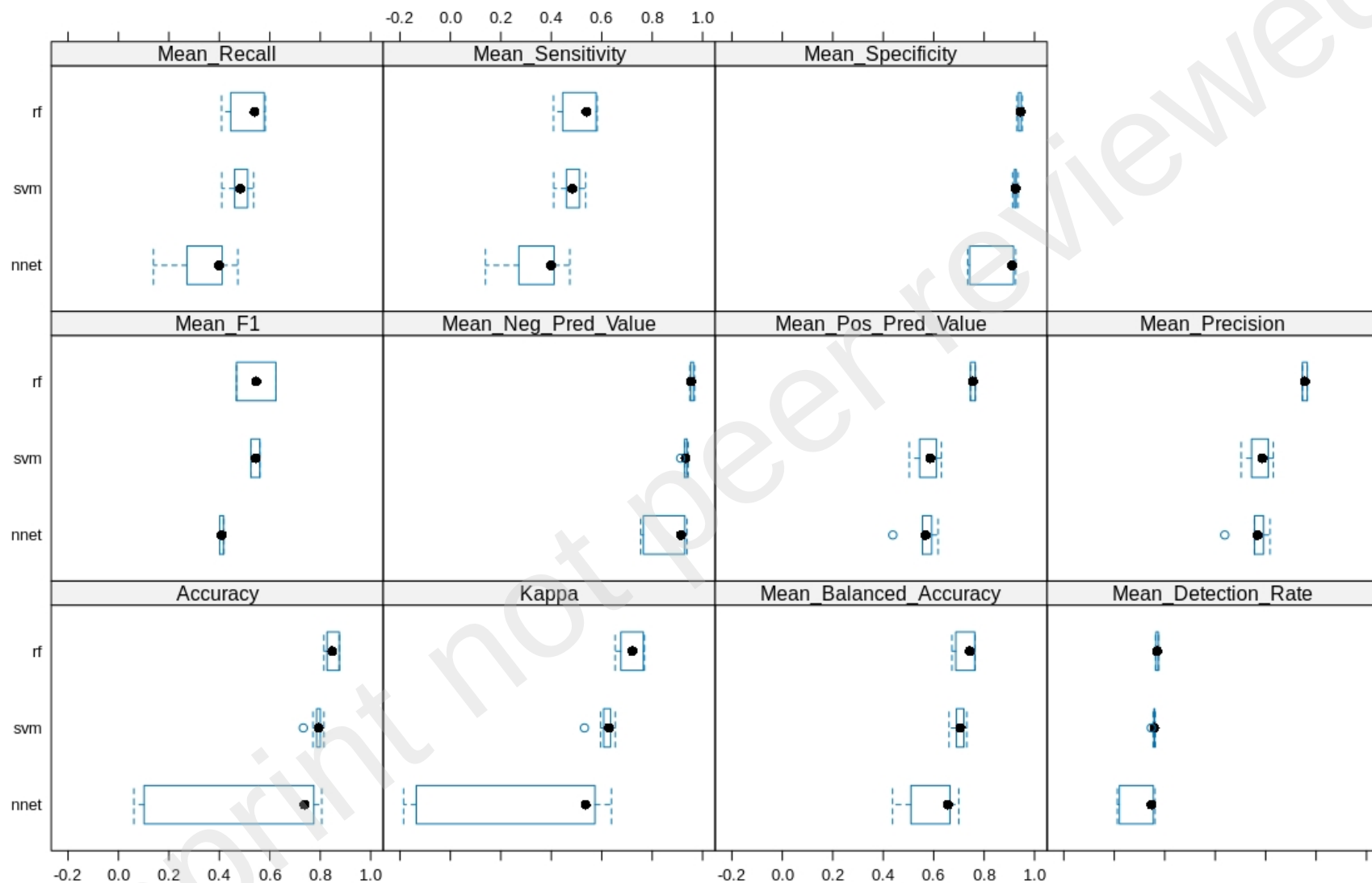
Digital surface model after removing non-ground components

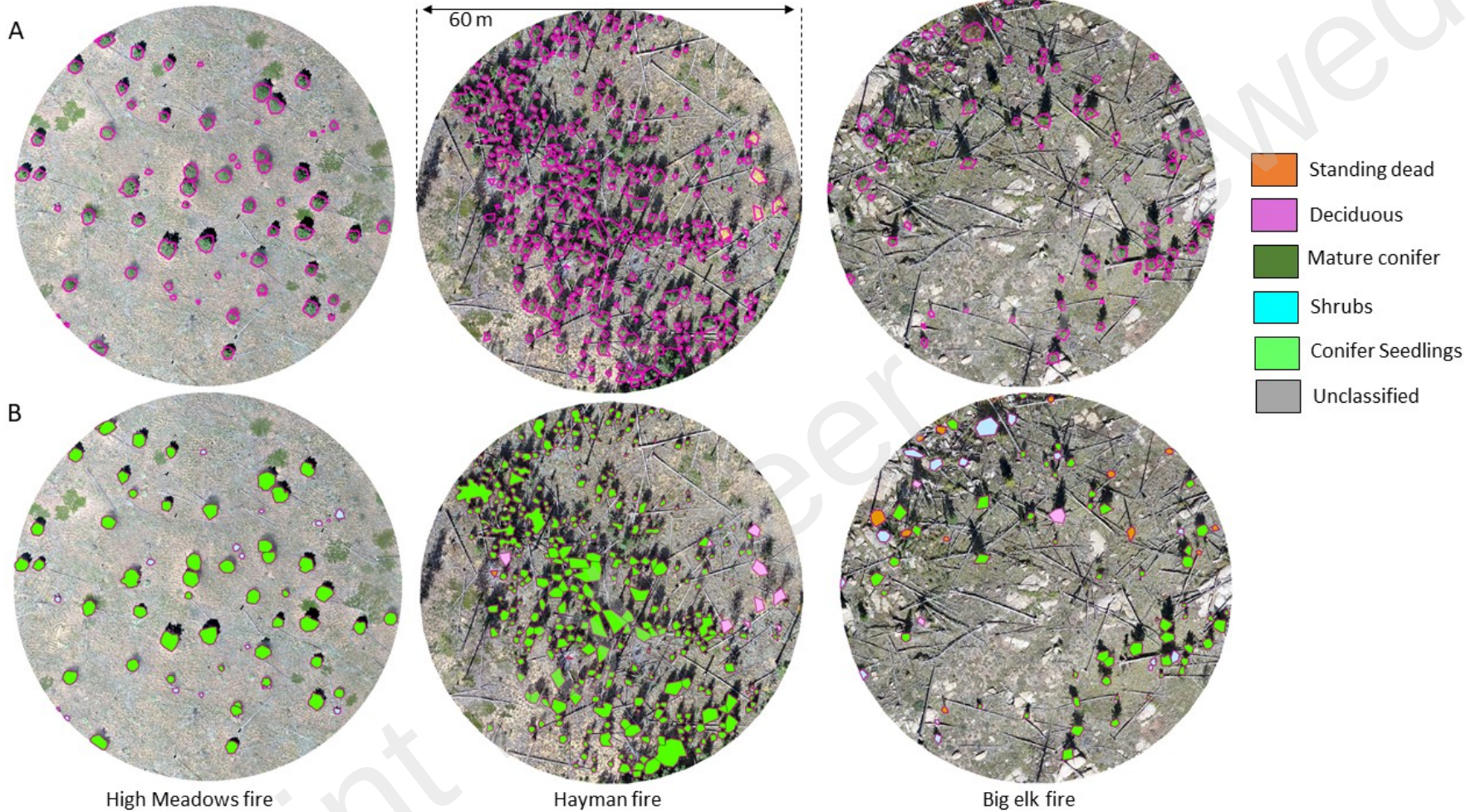


Individual vegetation locations as circular points on a canopy height model



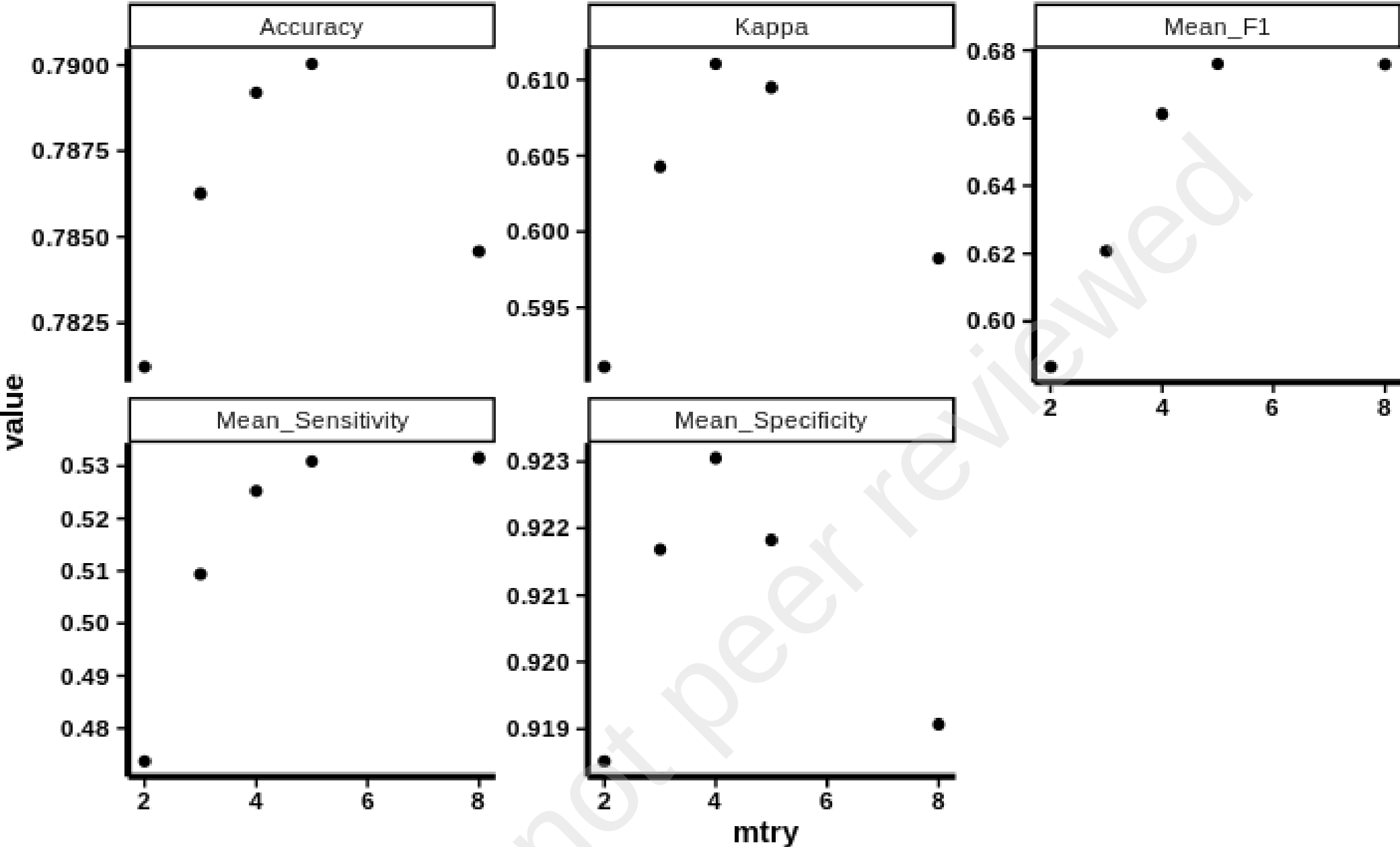






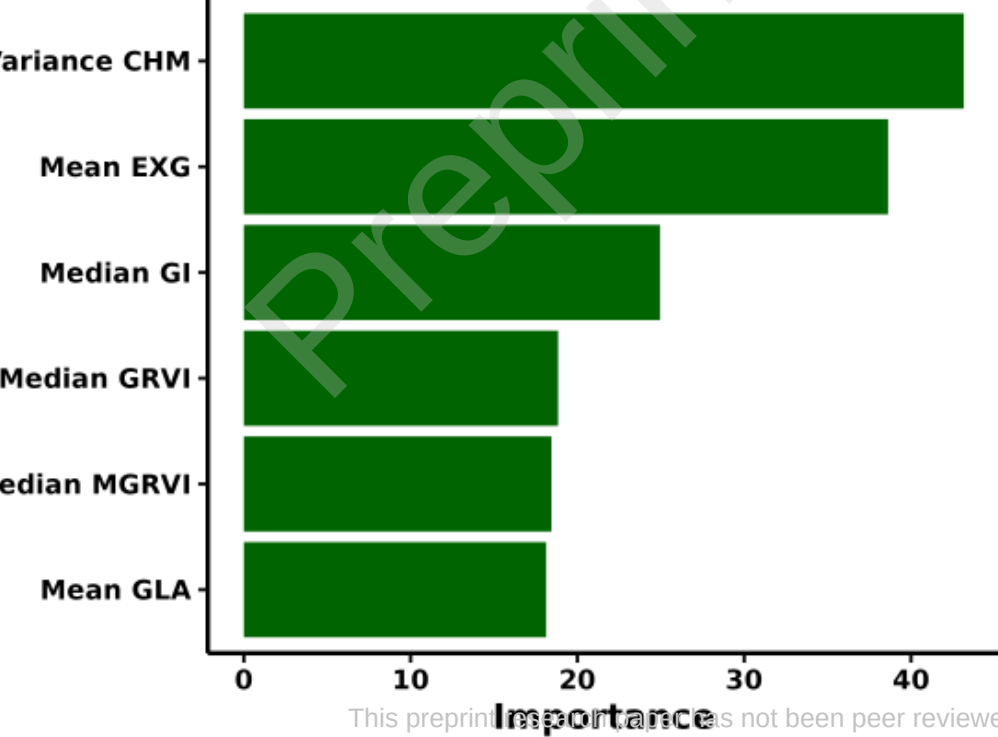
A

Model Performance



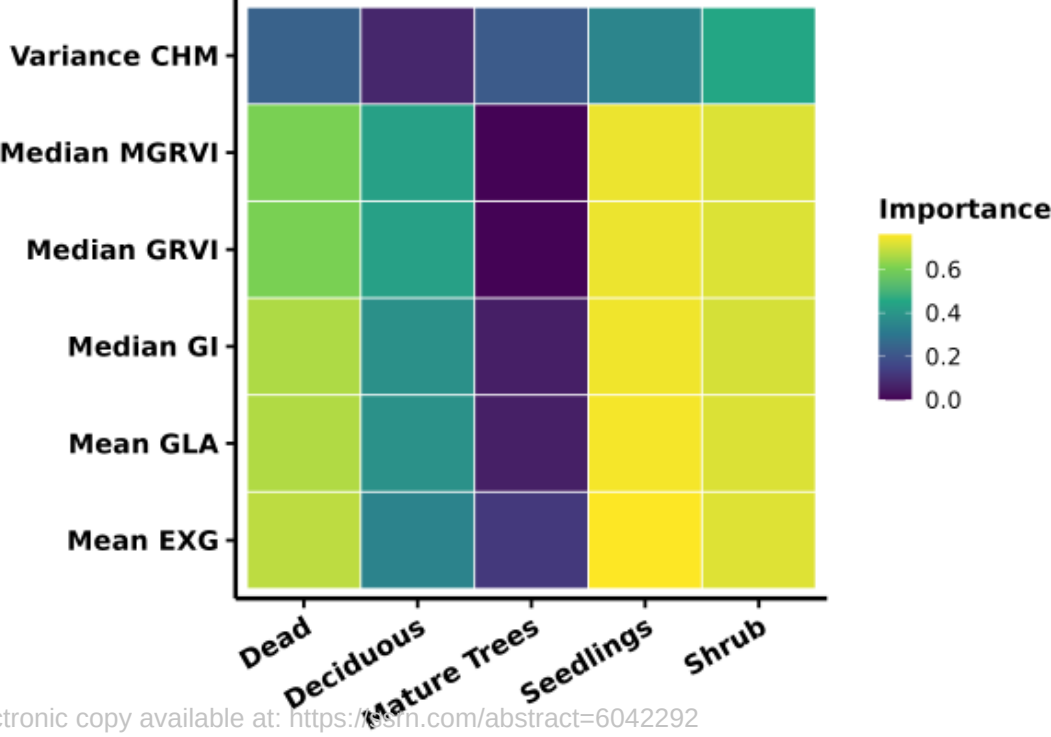
B

Variable Importance (Overall)



C

Class Importance Heatmap



Single-variable class separation based on AUC

